# msvtx_nMDTMiddleStation

overview · all regions and selections · rank 9 / 169



rank 9 / 169  by endcap / ht40 / D_equal

| region | selection | D_equal | D_pooled | 125/15 | 125/55 | 200/50 | 400/100 | 600/150 | 1000/275 | n_bkg |
|---|---|---|---|---|---|---|---|---|---|---|
| endcap | all | 0.648 | 0.736 | 0.519 | 0.504 | 0.579 | 0.738 | 0.789 | 0.827 | 34,970 |
| endcap | ht40 | 0.631 | 0.725 | 0.498 | 0.481 | 0.556 | 0.725 | 0.779 | 0.821 | 7,390 |
| endcap | bkg split | 0.050 | below 27,580 / above 7,390 | | | | | | | |
| barrel | all | 0.313 | 0.444 | 0.196 | 0.242 | 0.216 | 0.405 | 0.488 | 0.573 | 1,089 |
| barrel | ht40 | 0.379 | 0.510 | 0.221 | 0.306 | 0.282 | 0.472 | 0.554 | 0.643 | 220 |
| barrel | bkg split | 0.094 | below 869 / above 220 | | | | | | | |
| all | all | 0.483 | 0.573 | 0.355 | 0.321 | 0.405 | 0.568 | 0.608 | 0.670 | 36,059 |
| all | ht40 | 0.470 | 0.563 | 0.339 | 0.302 | 0.386 | 0.557 | 0.600 | 0.665 | 7,610 |
| all | bkg split | 0.046 | below 28,449 / above 7,610 | | | | | | | |

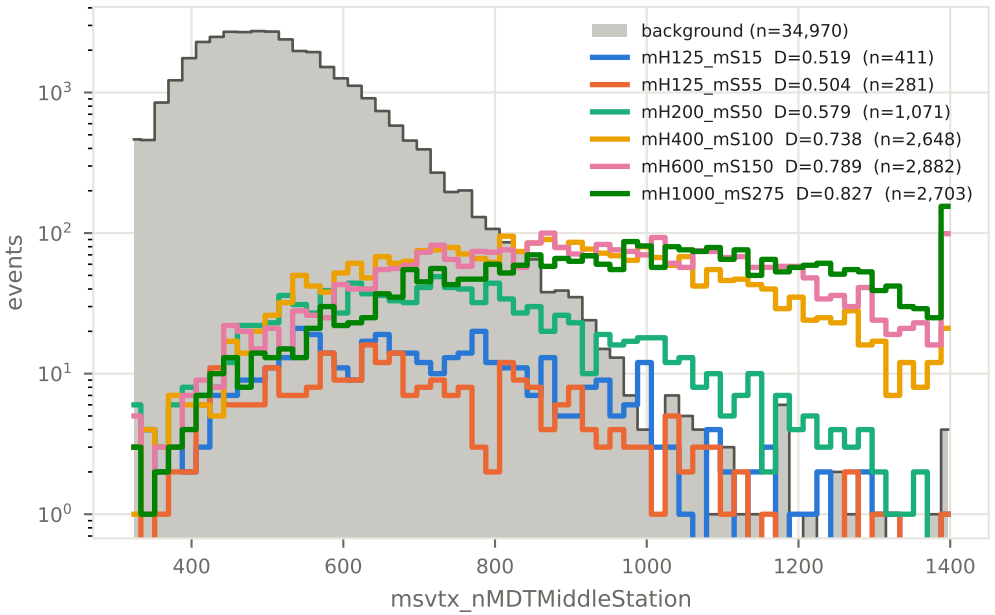
selections: all = every event;  ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

# msvtx_nMDTMiddleStation

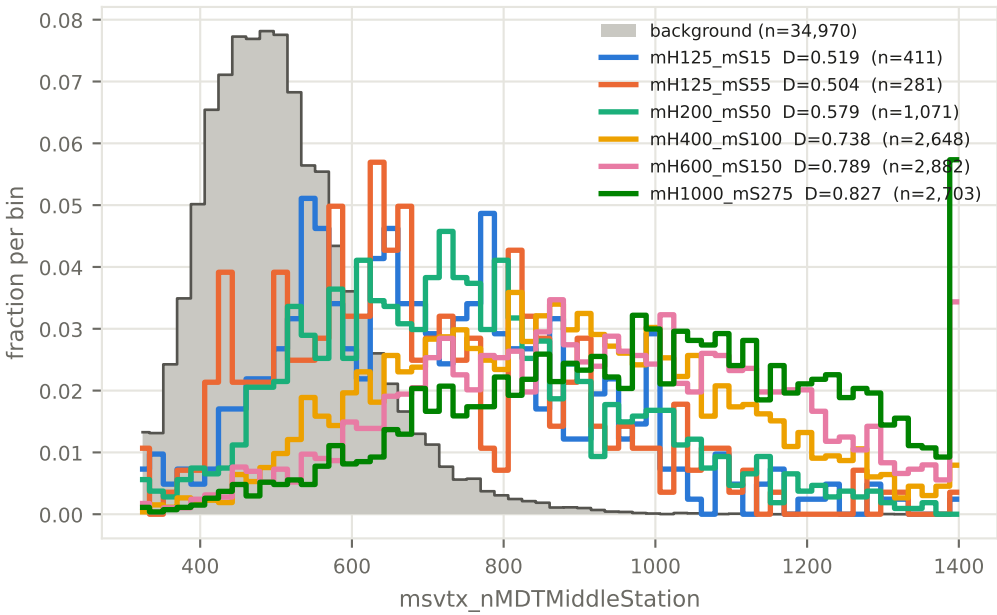
region: ENDCAP · selection: all (every event) · rank 9 / 169

D_equal 0.648 D_pooled 0.736 min 0.504 (mH125_mS55) max 0.827 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

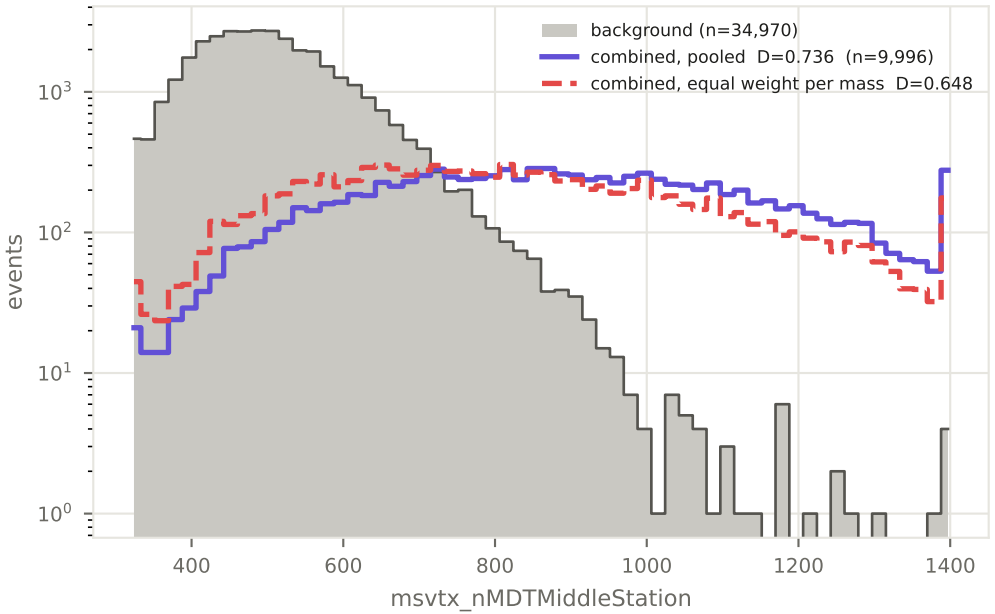
each mass point vs background · raw



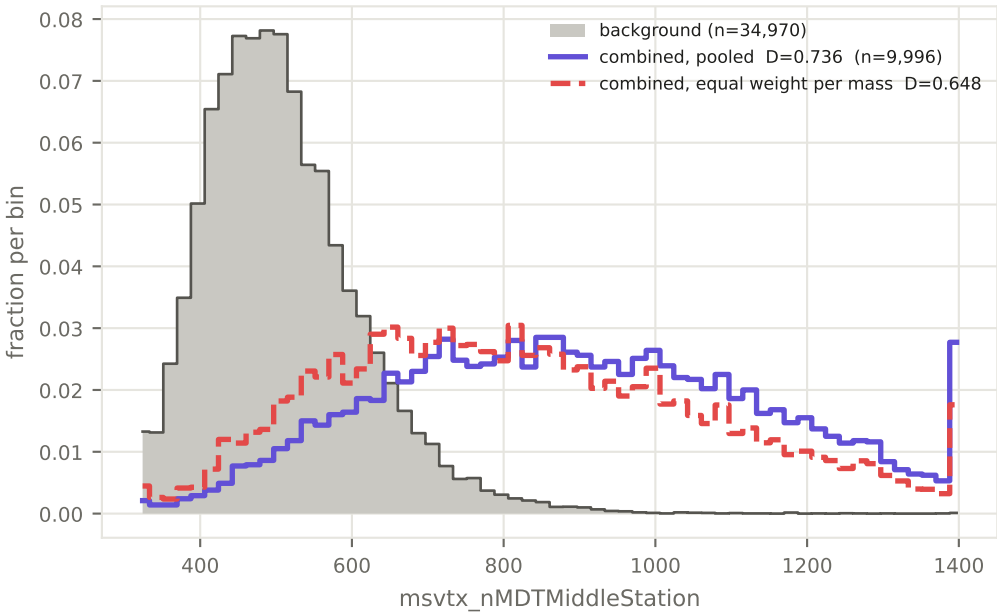

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized



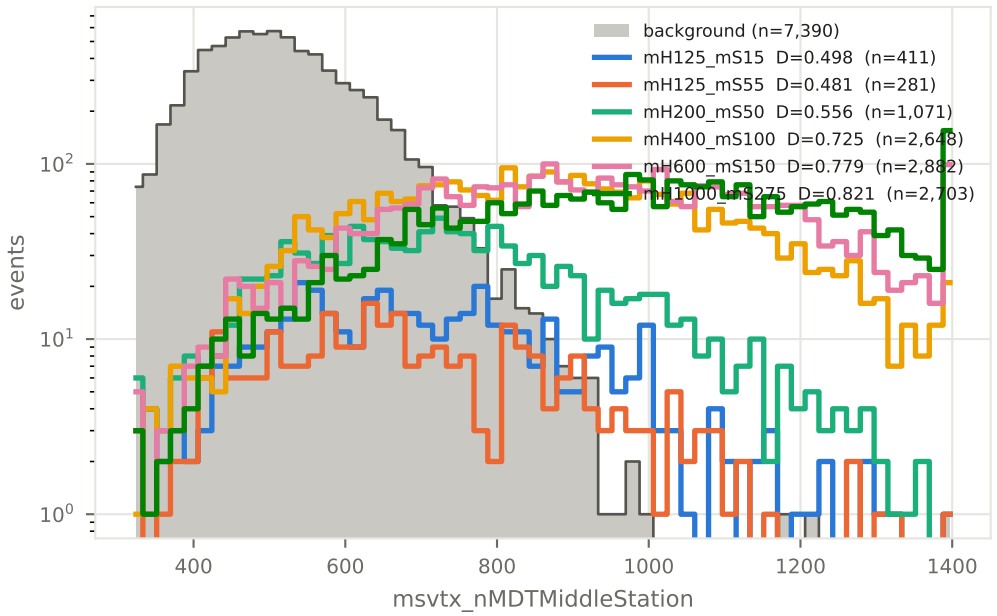

edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTMiddleStation

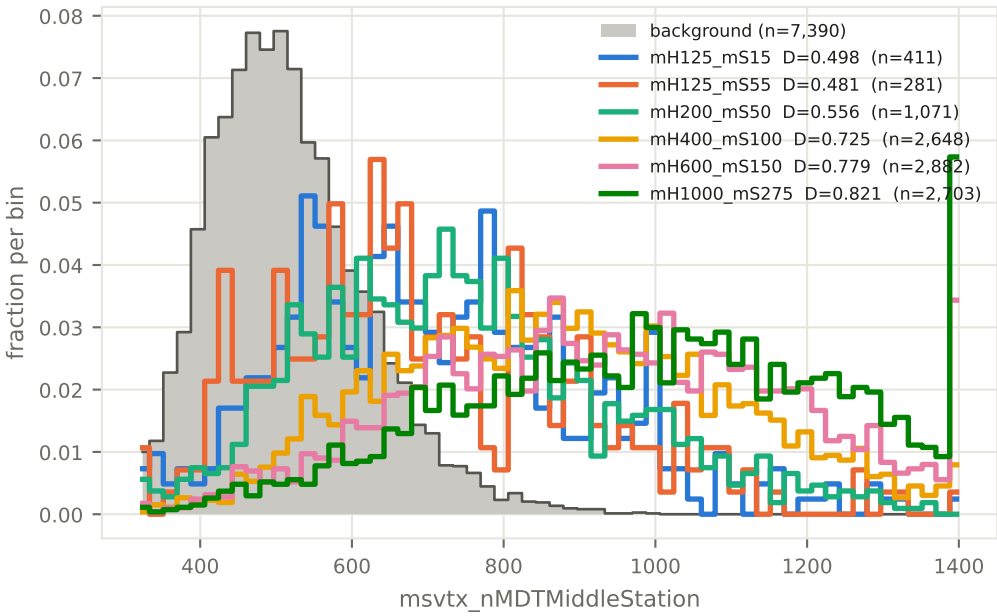
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 9 / 169

D_equal 0.631 D_pooled 0.725 min 0.481 (mH125_mS55) max 0.821 (mH1000_mS275) bkg n = 7,390 sig n = 9,996

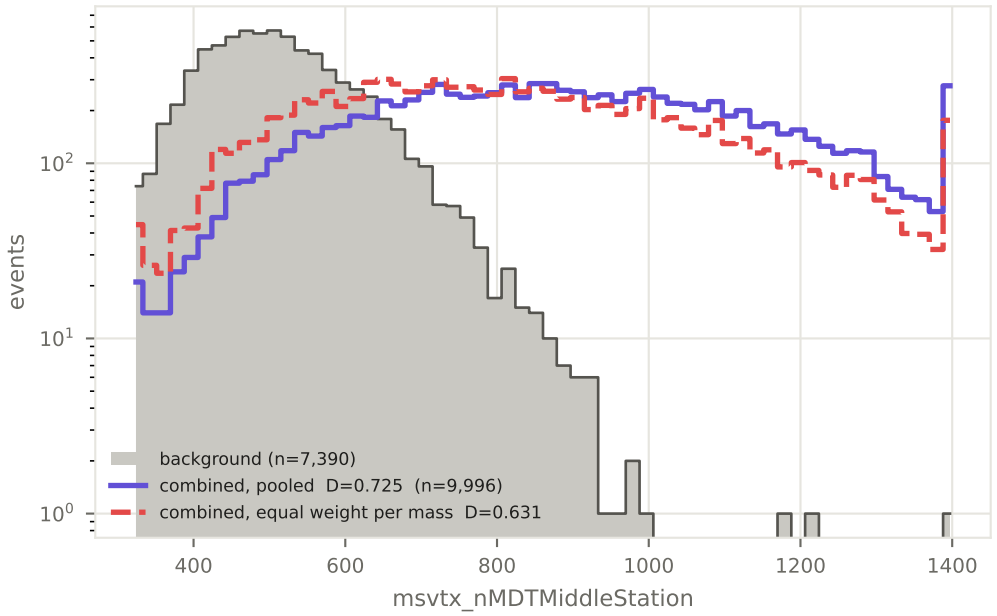
each mass point vs background · raw



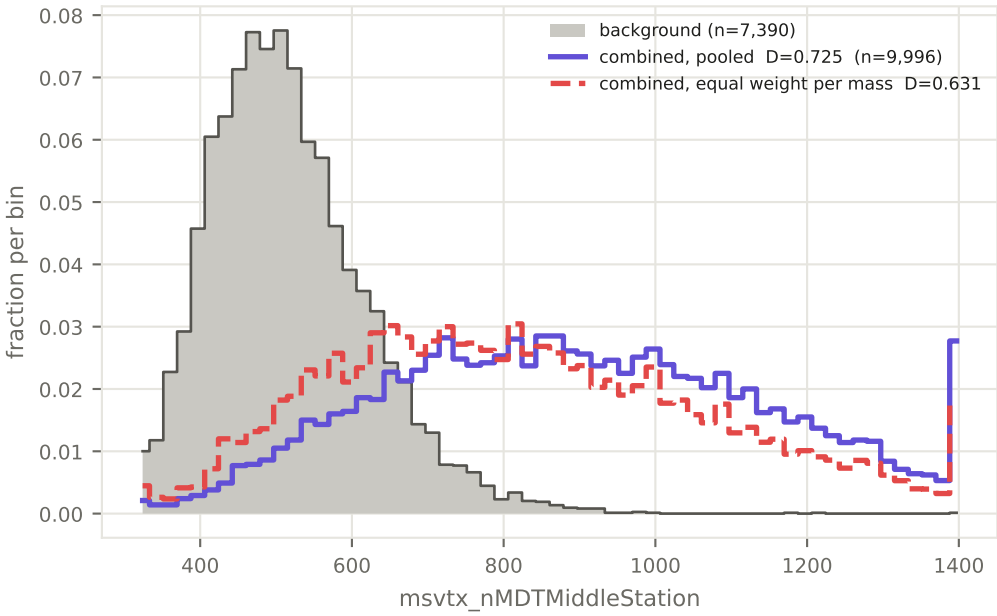

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized

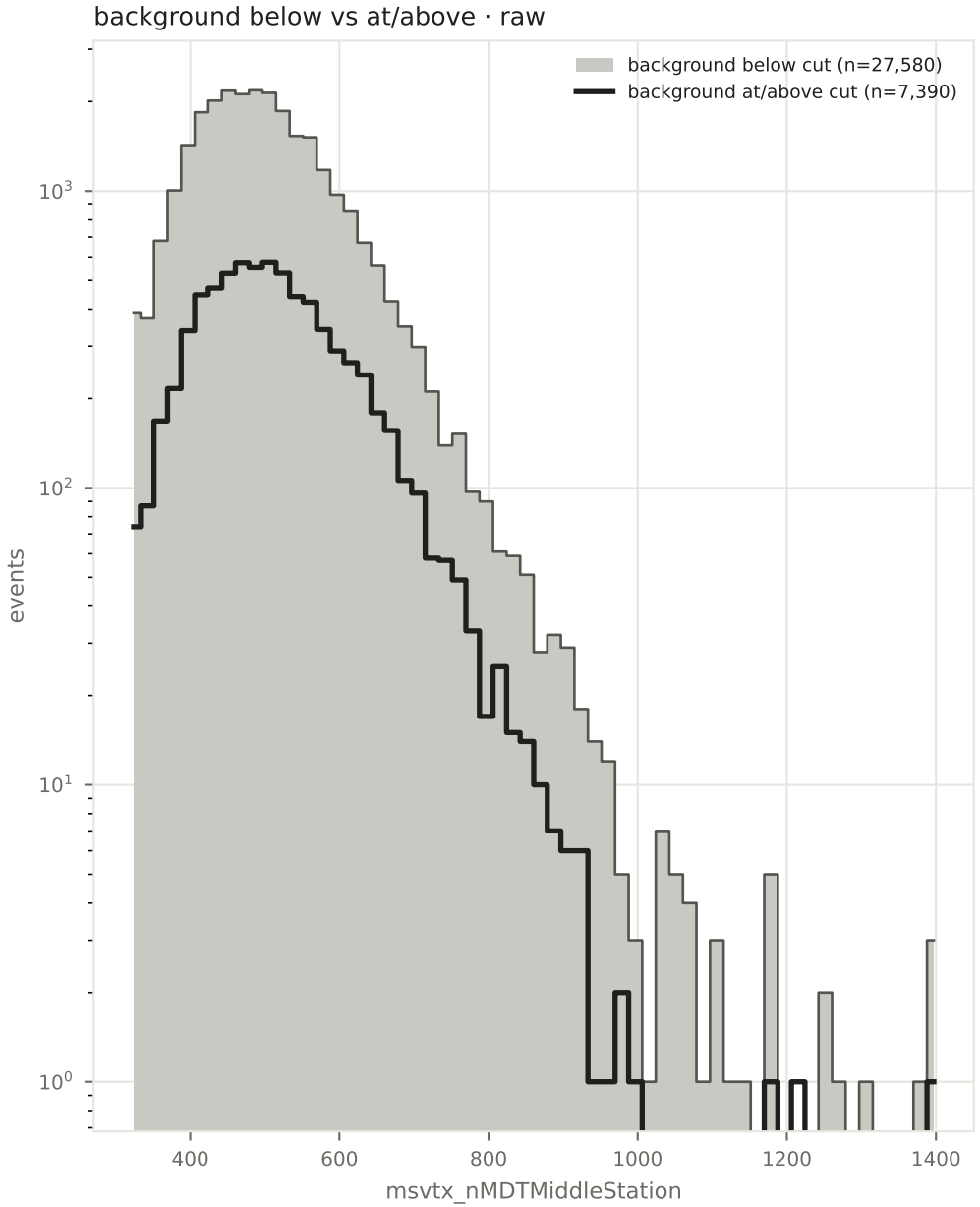


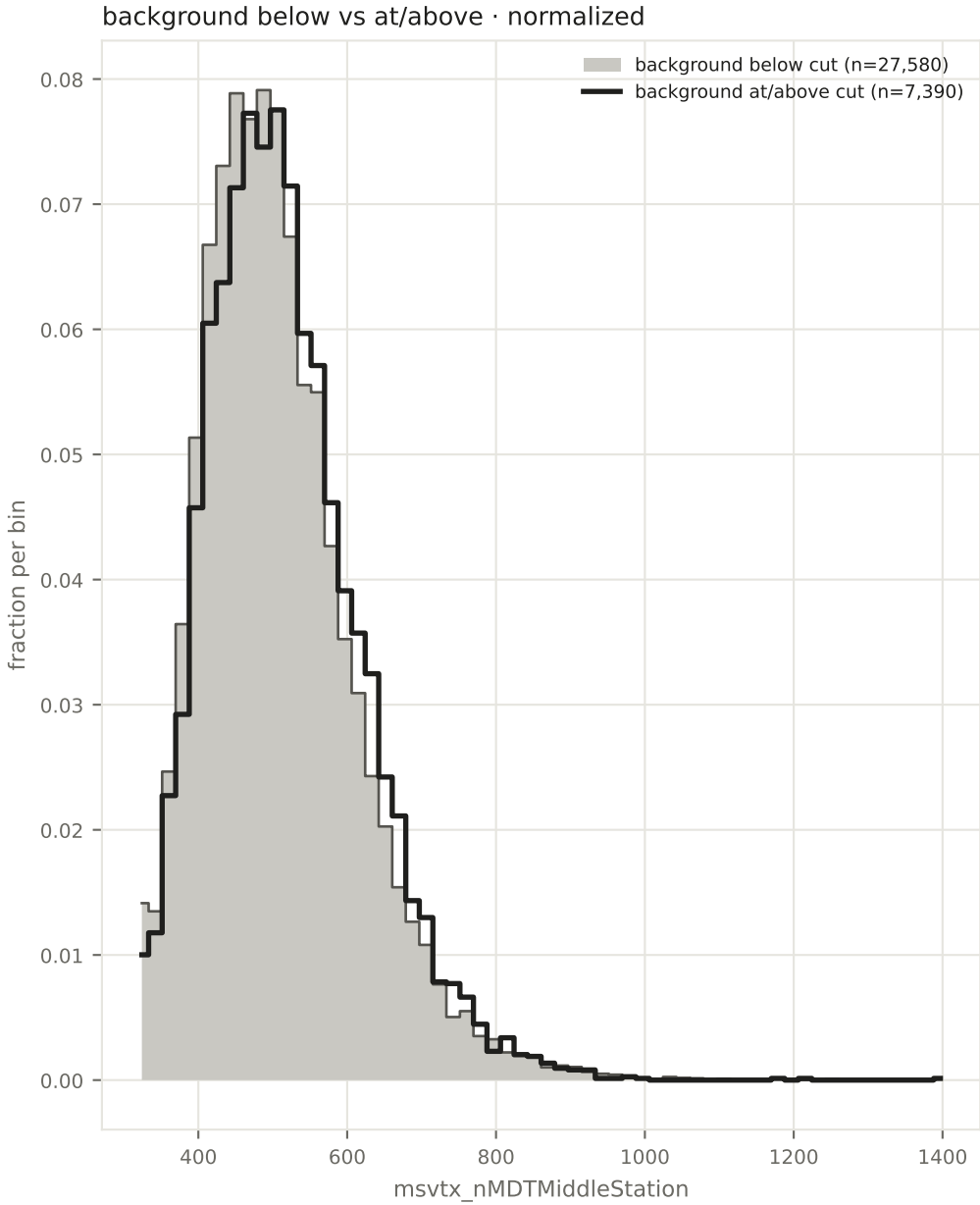

edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTMiddleStation

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 9 / 169

D(below vs above) 0.050  n below = 27,580  n above = 7,390  large D = background shape changes across the cut



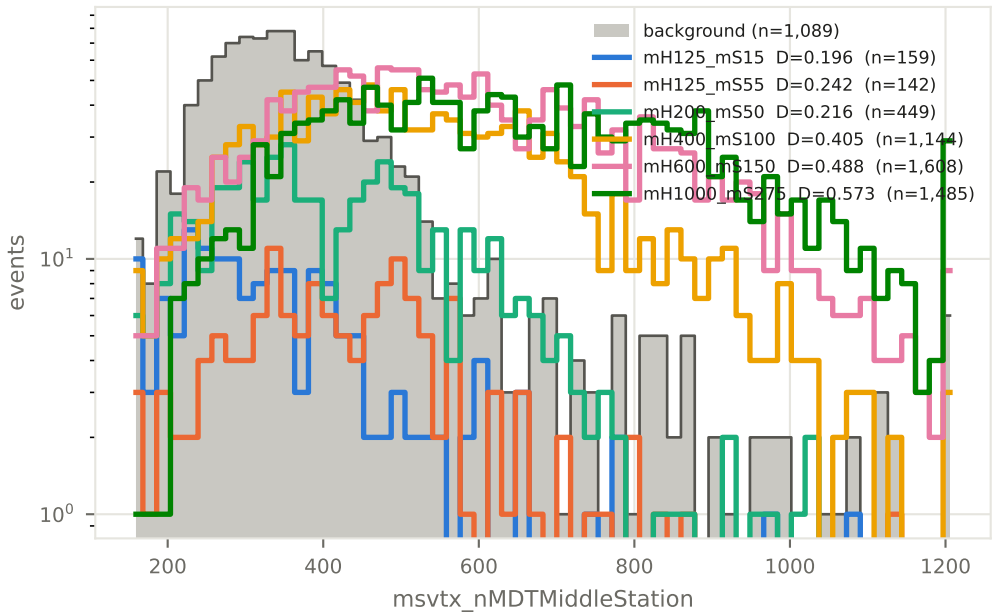



# msvtx_nMDTMiddleStation

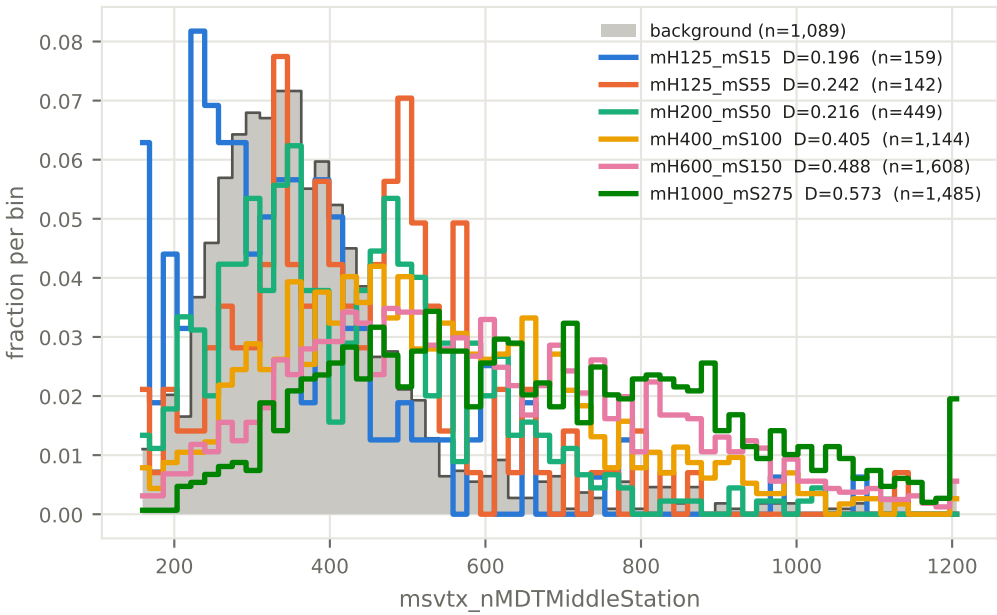
region: BARREL · selection: all (every event) · rank 9 / 169

D_equal 0.313 D_pooled 0.444 min 0.196 (mH125_mS15) max 0.573 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

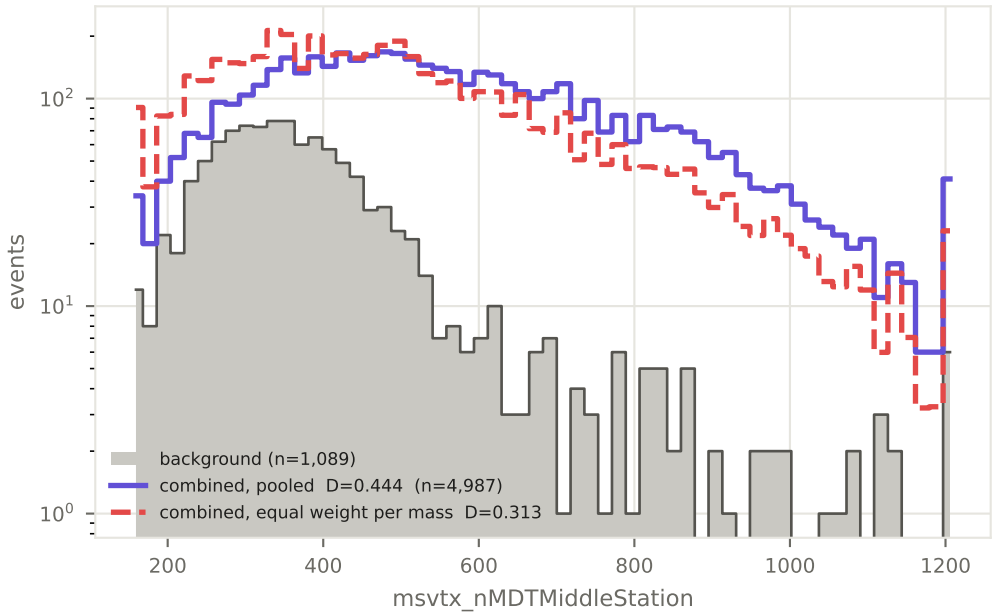
each mass point vs background · raw



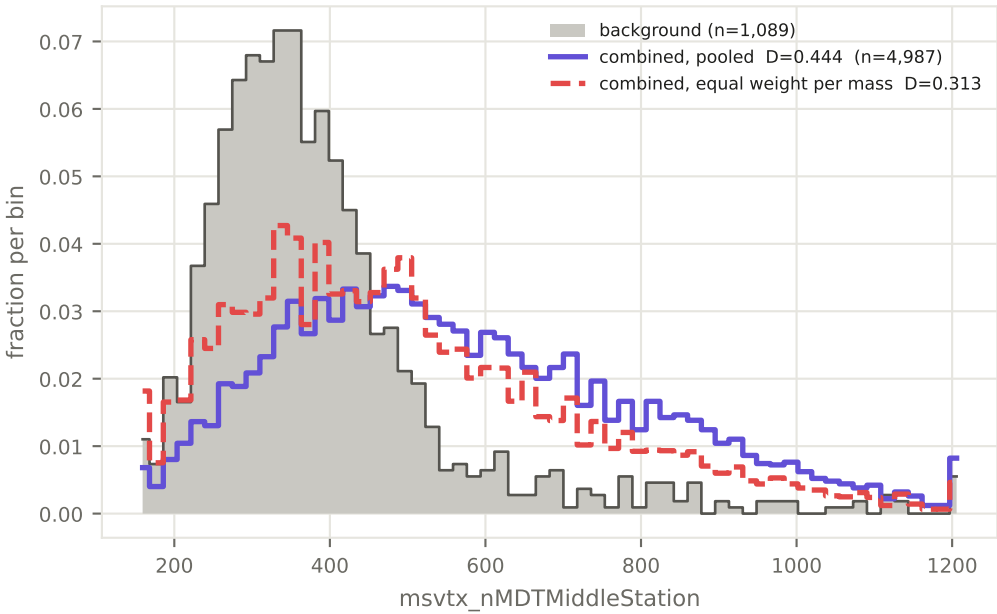

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized



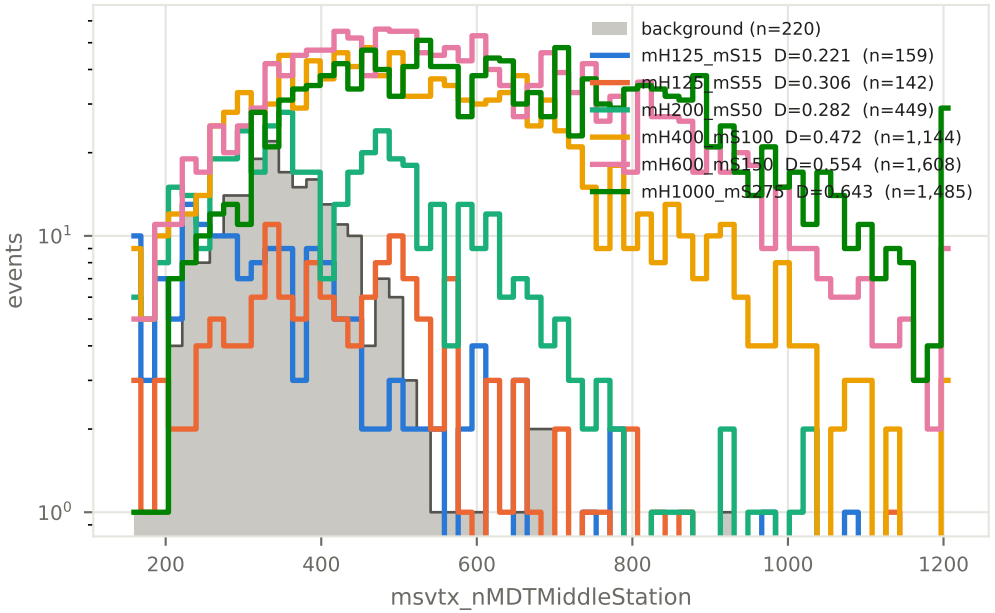

edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTMiddleStation

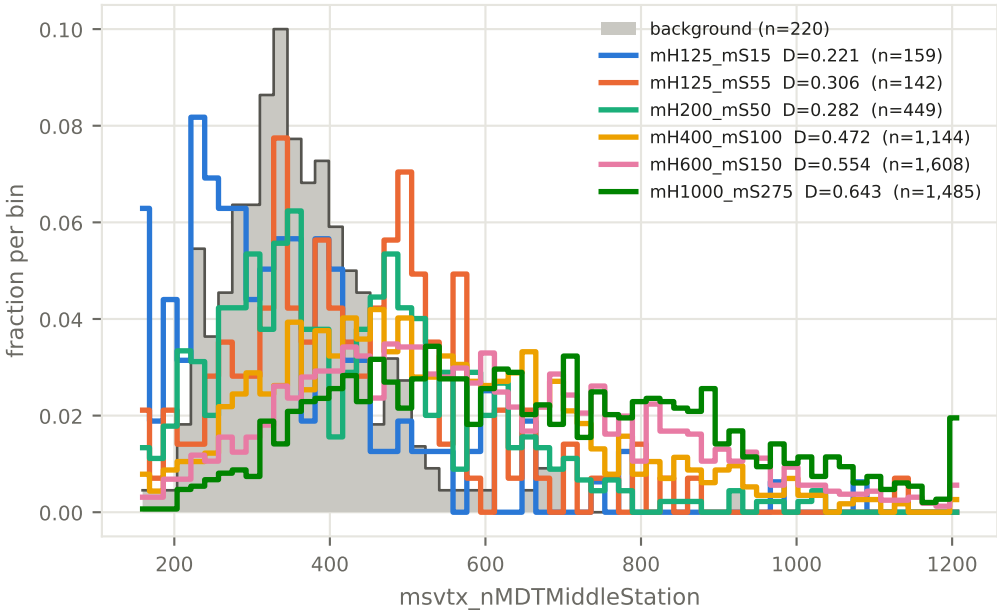
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 9 / 169

D_equal 0.379 D_pooled 0.510 min 0.221 (mH125_mS15) max 0.643 (mH1000_mS275) bkg n = 220 sig n = 4,987

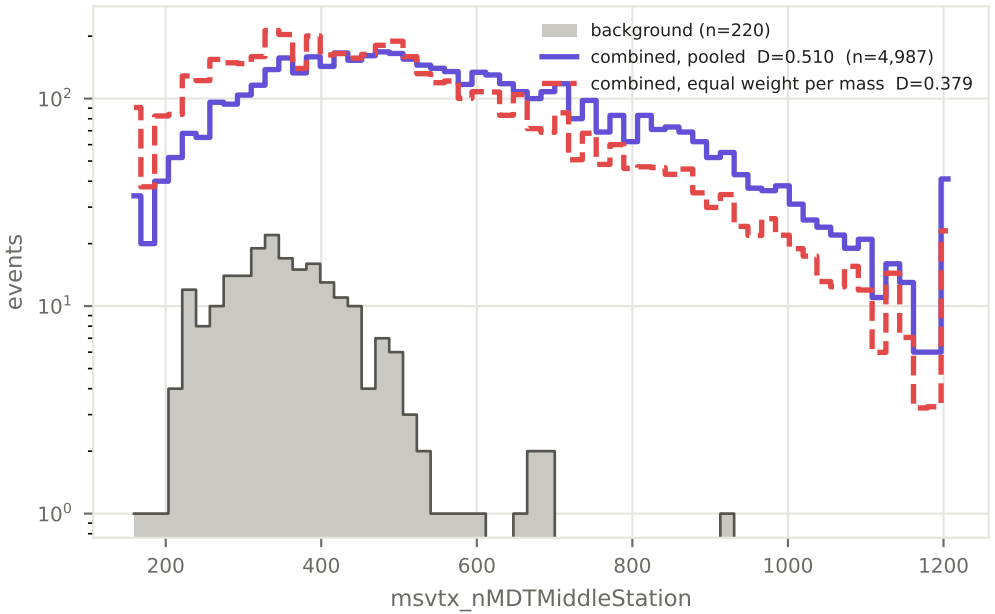
each mass point vs background · raw



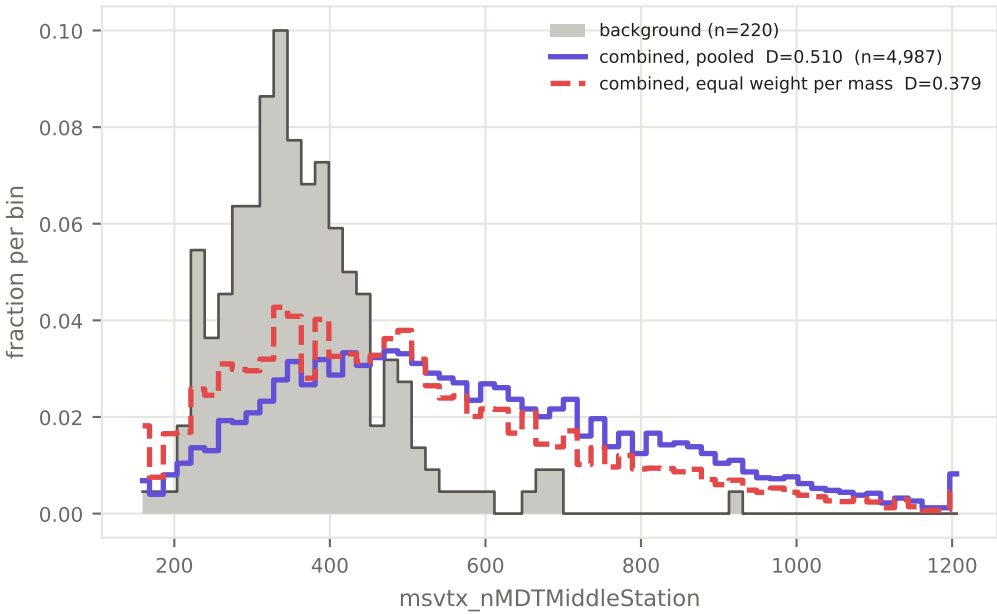

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized

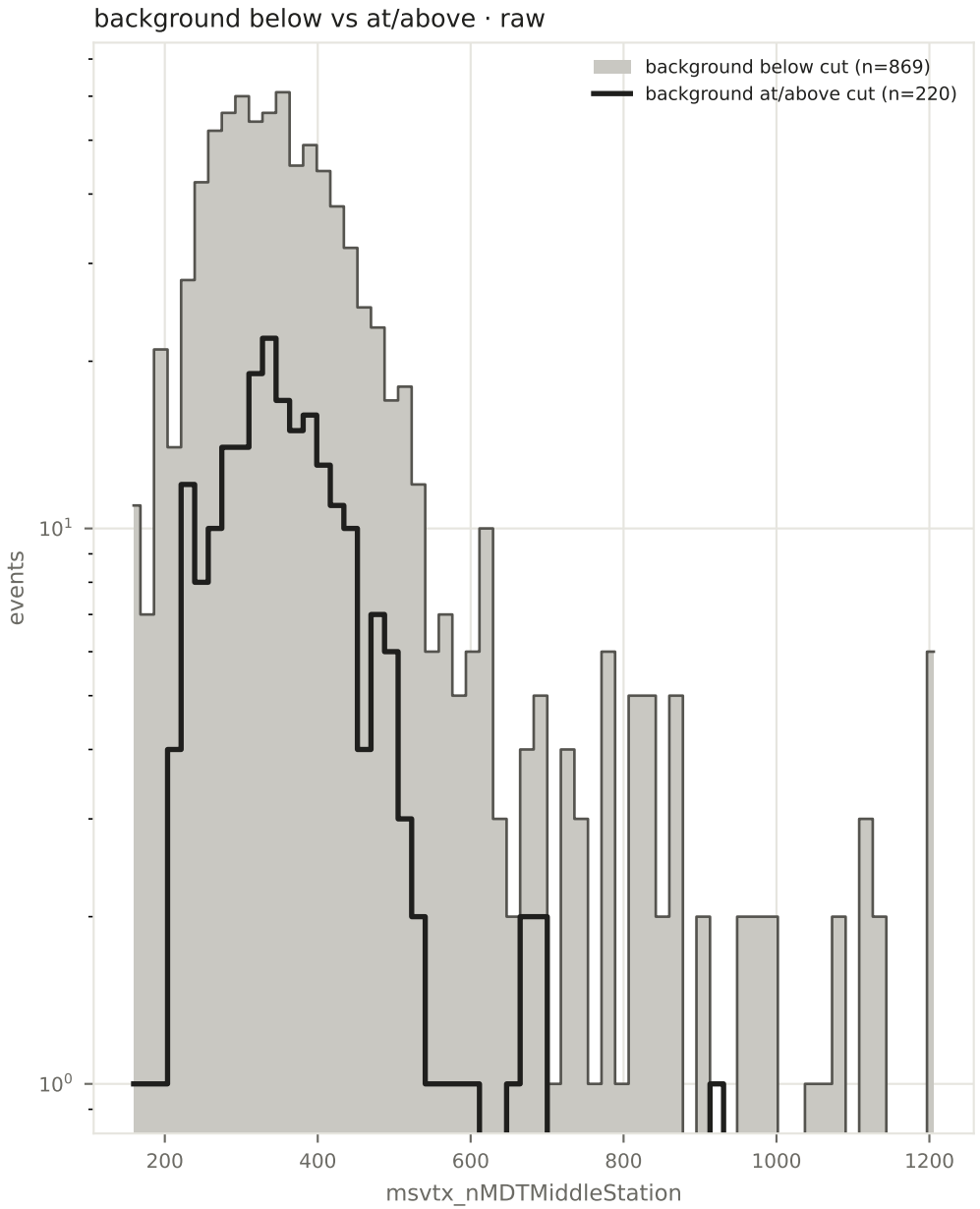


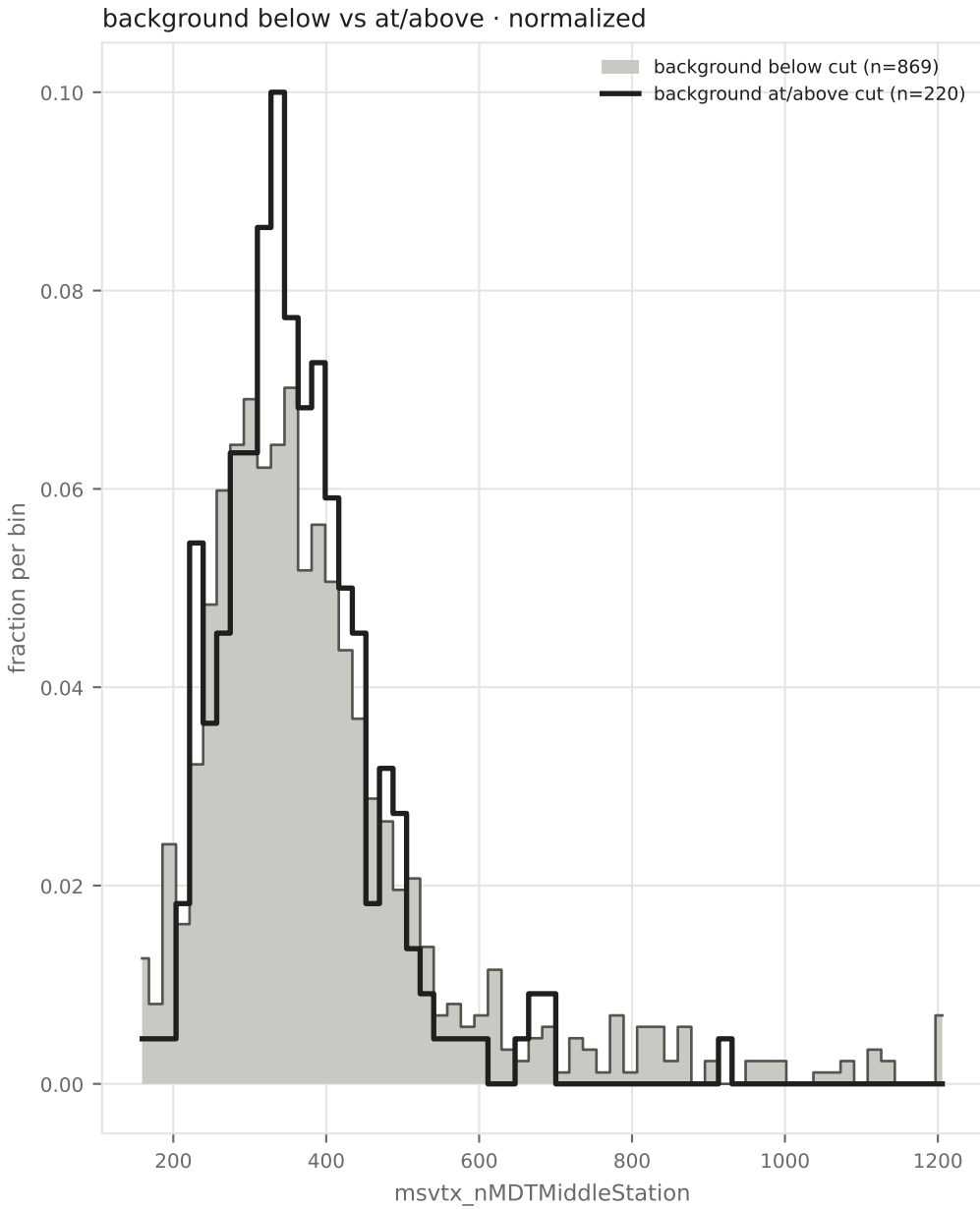

edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTMiddleStation

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 9 / 169

D(below vs above) 0.094  n below = 869  n above = 220  large D = background shape changes across the cut



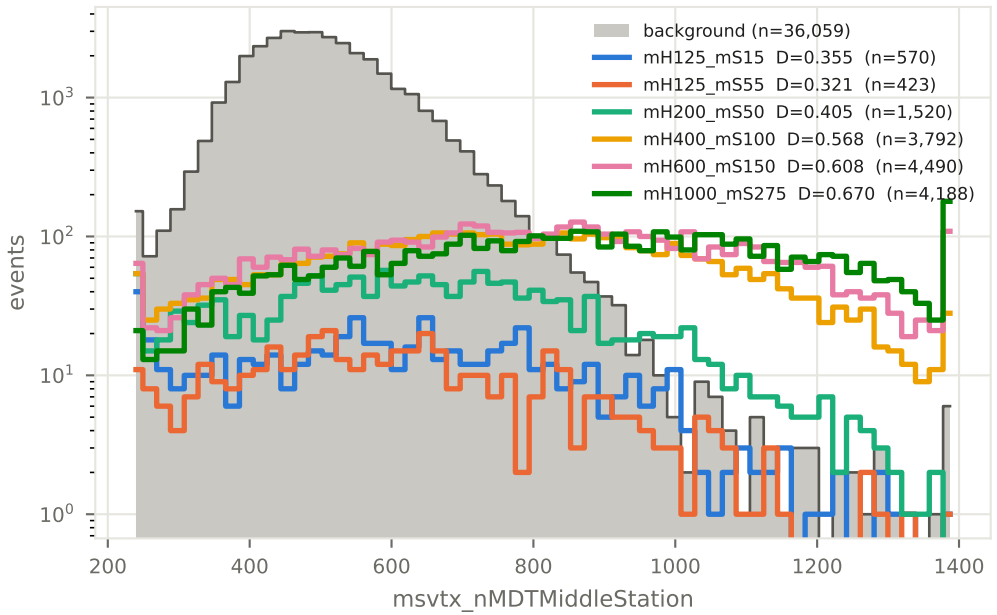



# msvtx_nMDTMiddleStation

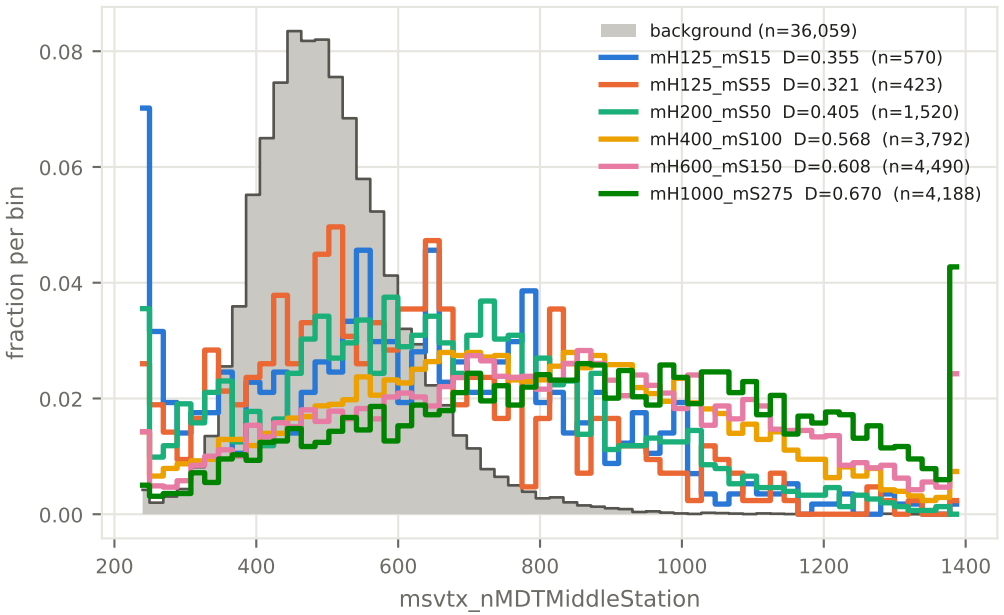
region: ALL · selection: all (every event) · rank 9 / 169

D_equal 0.483 D_pooled 0.573 min 0.321 (mH125_mS55) max 0.670 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

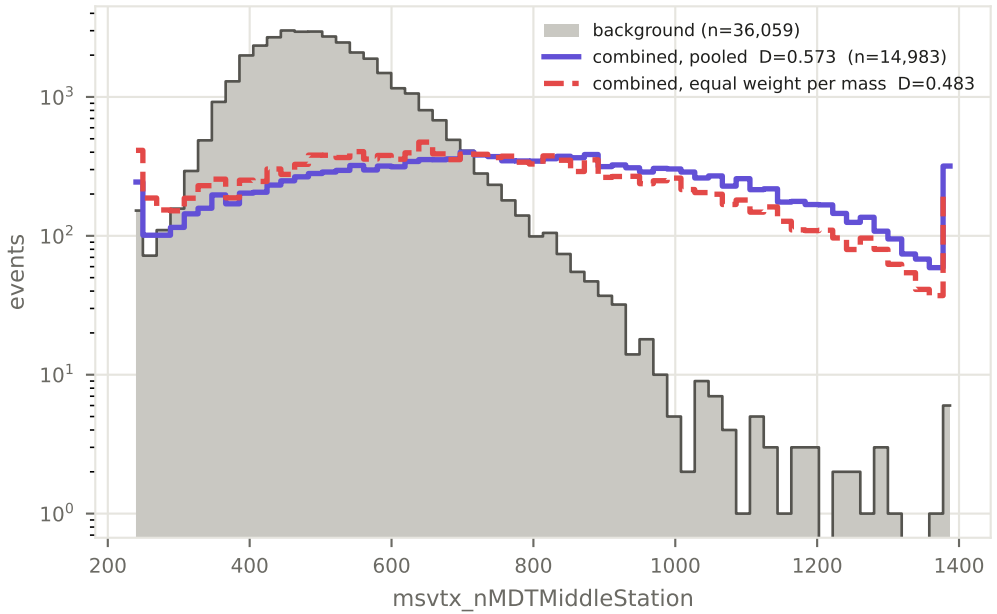
each mass point vs background · raw



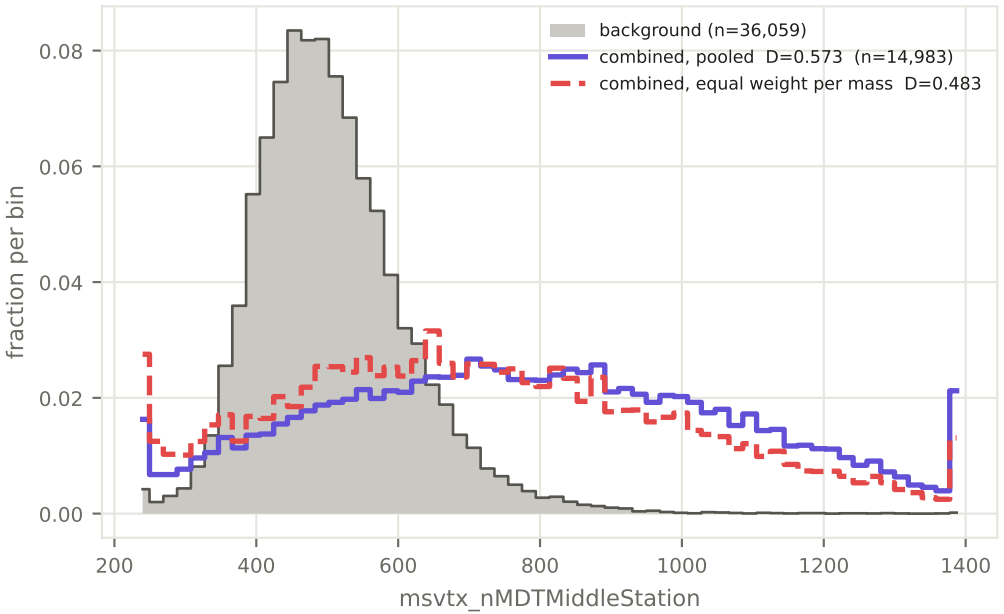

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized



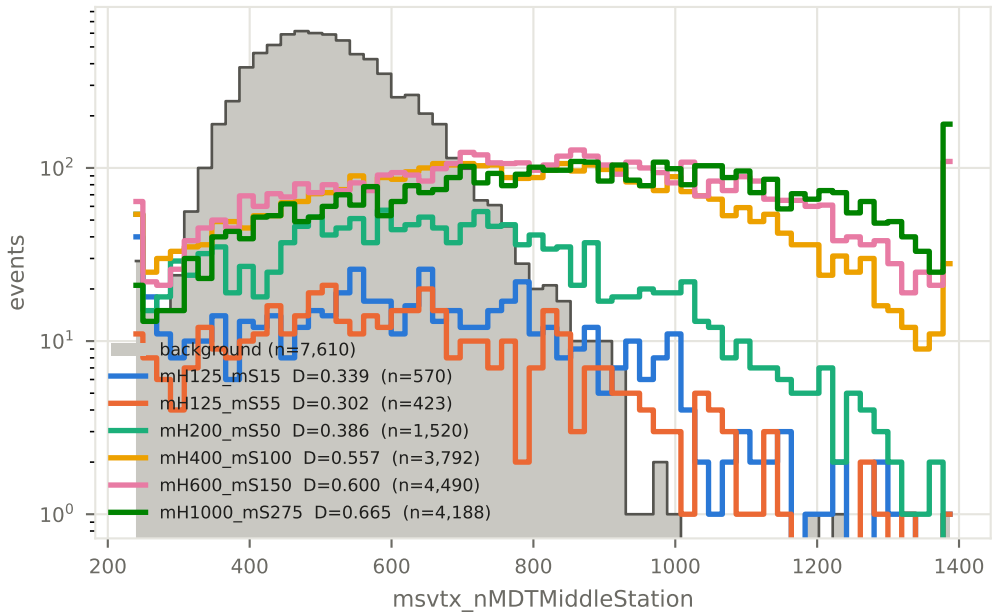

edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTMiddleStation

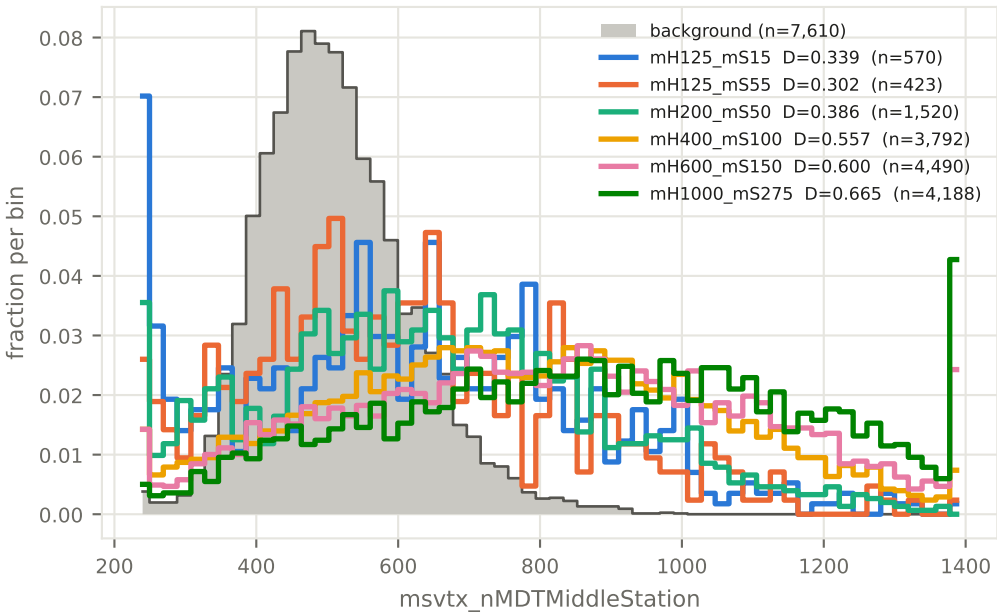
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 9 / 169

D_equal 0.470 D_pooled 0.563 min 0.302 (mH125_mS55) max 0.665 (mH1000_mS275) bkg n = 7,610 sig n = 14,983

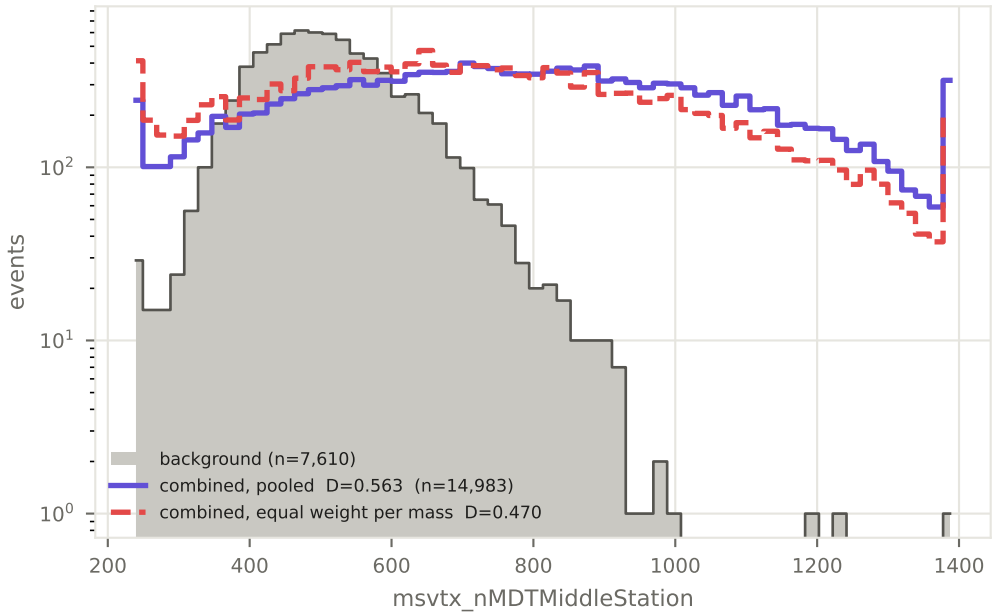
each mass point vs background · raw



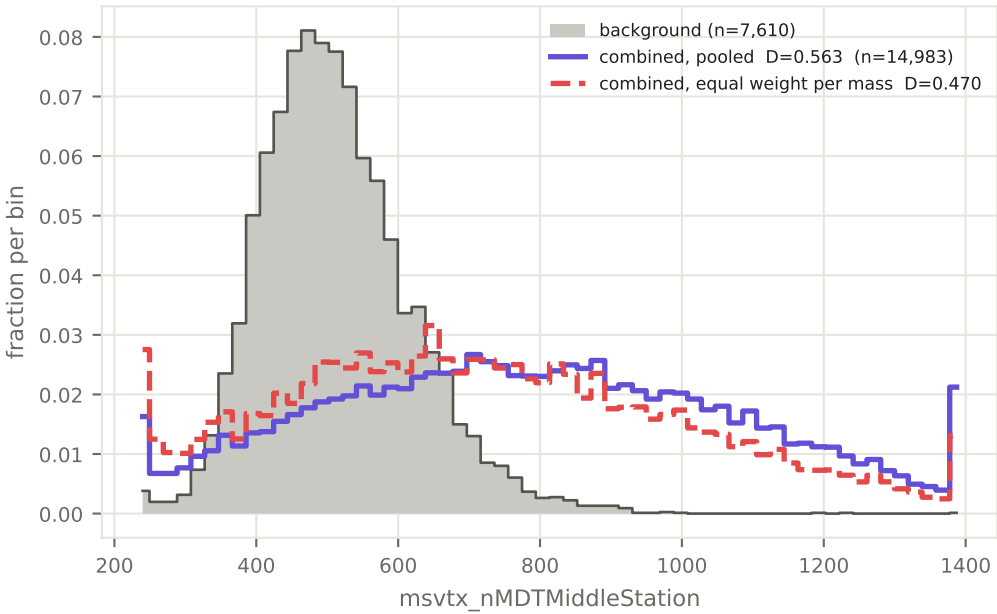

each mass point vs background · normalized




combined signal vs background · raw




combined signal vs background · normalized




edge bins include under/overflow · NaN and -999 removed

# msvtx_nMDTMiddleStation

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 9 / 169

D(below vs above) 0.046  n below = 28,449  n above = 7,610  large D = background shape changes across the cut

